

A Novel Dual-stream Residual Fusion Network (DRF-Net) for Accurate Detection of Face Morphing Attack in Biometric Systems

Abstract—Face recognition systems are widely used in biometric security applications. However, their susceptibility to face morphing attacks poses a serious threat to identity verification. Morphing attacks generate highly realistic composite facial images that can evade both human inspection and automated recognition systems, motivating the need for robust morphing attack detection (MAD). Existing detection methods based on handcrafted features or conventional deep learning models often fail to capture subtle morphing artifacts, particularly when global facial structure is preserved. To address this gap, this research proposes DRF-Net, a dual-stream residual fusion network designed for effective face morphing attack detection. The proposed approach employs parallel encoder-decoder architectures to jointly learn semantic facial representations and reconstruction-based residual noise, explicitly modeling morphing-induced inconsistencies that are suppressed in the spatial domain. Feature-level fusion of semantic and residual representations enhances sensitivity to fine-grained morphing cues. Experimental results on a large-scale dataset demonstrate that DRF-Net achieves superior performance compared to state-of-the-art methods, obtaining a BPCER of 0.0156 and an EER of 0.0119. These results highlight the effectiveness of residual-aware dual-stream learning for robust face morphing attack detection in security-critical biometric systems.

Index Terms—Face morphing attack detection, biometric security, residual noise modeling, dual-stream deep learning, encoder-decoder networks.

I. INTRODUCTION

Biometric authentication has become an integral component of modern identity verification systems, including border control, passport issuance, surveillance, and digital identity management. Among various biometric modalities, face recognition systems (FRS) are widely adopted due to their contactless operation, ease of deployment, and high user acceptance. Despite these advantages, the increasing reliance on FRS has also exposed critical security vulnerabilities, particularly to sophisticated presentation attacks that can undermine system reliability. One of the most serious threats to FRS is the face morphing attack, where facial characteristics of two or more individuals are digitally blended to generate a single composite image. When such a morphed image is enrolled as an identity document photograph, it can successfully match multiple individuals during verification, enabling identity fraud. Morphing attacks are especially dangerous in high-security scenarios, as morphed images often appear visually realistic and are difficult to distinguish from bona fide samples. Previous studies have demonstrated that even state-of-the-art face recognition systems exhibit high vulnerability to morphing attacks, partic-

ularly under operational conditions [1]. Furthermore, recent advances in deep generative models have significantly improved the visual quality of morphed images, further reducing the effectiveness of both human inspection and traditional detection mechanisms. Detecting face morphing attacks remains a challenging problem due to the subtle nature of morphing artifacts. Early detection approaches primarily relied on handcrafted texture descriptors and signal-level features to identify inconsistencies introduced during the morphing process. While these methods can capture certain local texture variations, their robustness is limited when confronted with high-quality morphs or real-world variations such as compression, illumination changes, and post-processing effects [2]. Deep learning-based approaches have shown improved performance by learning discriminative facial representations; however, models that operate solely in the spatial domain may fail to detect morphing attacks when global facial structure is well preserved. To address these limitations, residual noise analysis has emerged as an effective complementary strategy for morphing attack detection. Residual representations, obtained by subtracting reconstructed images from their original counterparts, emphasize subtle inconsistencies introduced by blending, warping, and smoothing operations during morph generation. These artifacts are often suppressed in appearance-based features but become more distinguishable in the residual domain, making residual learning particularly suitable for detecting high-quality morphing attacks.

Motivated by these observations, this work proposes a dual-stream residual-aware face morphing attack detection framework, termed DRF-Net, which integrates handcrafted texture descriptors with deep encoder-decoder architectures. The proposed framework employs parallel deep learning streams to learn complementary semantic and forensic representations, while explicitly modeling reconstruction-based residual noise. By fusing semantic features and residual artifacts at the feature level, the proposed approach enhances sensitivity to subtle morphing cues and improves robustness under challenging attack scenarios. The key contributions of this work are summarized as follows:

- A dual-stream residual fusion network (DRF-Net) is proposed for face morphing attack detection, which jointly learns semantic facial features and reconstruction-based residual representations to enhance sensitivity to subtle morphing artifacts.

- The proposed framework integrates handcrafted texture descriptors with deep encoder-decoder architectures, enabling robust detection of high-quality morphs under realistic variations.
- Extensive experimental evaluations demonstrate that the proposed approach achieves lower error rates compared to existing methods, validating its effectiveness for reliable face morphing attack detection.

The rest of this paper is structured as follows. A survey of the relevant works is given in Section II. The general foundation for morph detection is shown in Section III. The performance assessment and experimental outcomes of the suggested model are covered in Section IV. The paper is finally concluded and the key findings are summarized in Section V.

II. RELATED WORKS

Face morphing attacks pose a serious threat to face recognition systems, as they enable multiple identities to be verified using a single composite facial image. A morphed face is typically generated by blending two or more genuine facial images in a manner that preserves visual realism while deceiving automated recognition systems. Consequently, recent research has focused on developing robust detection methods using deep learning, residual noise analysis, and image quality assessment to enhance biometric security. Deep learning-based approaches have gained significant attention for face morphing attack detection due to their ability to learn discriminative facial representations. Convolutional neural networks (CNNs), autoencoders (AEs), and generative adversarial networks (GANs) have been widely explored for detecting morphing artifacts [3]–[5]. These methods are typically trained on diverse datasets to improve robustness against varying morphing techniques and image qualities [6], [7]. However, deep models relying solely on spatial information often struggle to generalize to unseen morphing methods [8]. Residual noise analysis has emerged as an effective complementary strategy for morphing attack detection. Morphing operations introduce subtle noise inconsistencies during blending and warping, which are often imperceptible in the spatial domain. Raja *et al.* [9] proposed a generalized residual learning framework that demonstrated strong detection performance across datasets. Similarly, Venkatesh *et al.* [3] introduced a multi-scale context aggregation network to capture fine-grained residual artifacts, achieving improved detection accuracy at the cost of increased computational complexity. Deep color residual noise analysis was further explored in [10], highlighting the effectiveness of residual representations for morph detection, particularly in color-sensitive scenarios. Comprehensive surveys have provided valuable insights into the state of face morphing attack detection research. Venkatesh *et al.* [7] presented an extensive survey covering attack generation, detection techniques, datasets, and evaluation protocols, identifying key challenges such as limited generalization and lack of standardized benchmarks. Scherhag *et al.* [11] systematically analyzed the vulnerability of face recognition systems to morphing attacks, emphasizing the urgent need for reliable detection mechanisms.

Texture-based and sensor-level analysis methods have also been investigated for morphing detection. Techniques based on Photo Response Non-Uniformity (PRNU) and Fourier Spectrum of Sensor Pattern Noise (FS-SPN) analyze micro-level noise patterns introduced during morphing [12], [13].

While these methods provide additional security by exploiting sensor artifacts, their effectiveness may degrade under image compression and cross-sensor scenarios. Recent works have explored advanced topics such as GAN-generated morphs, identity disentanglement, and fairness in morphing attack detection. Venkatesh *et al.* [14] demonstrated that GAN-generated morphs pose threats comparable to traditional landmark-based morphs.

Overall, existing approaches face challenges related to generalization, computational efficiency, and robustness to unseen attacks. These limitations motivate the development of hybrid detection frameworks that combine deep learning with residual noise analysis, as pursued in this research.

III. METHODOLOGY

This section introduces the overall framework for the morph detection. The upcoming sub-sections include data description, data pre-processing, the proposed dual-stream residual fusion architecture, and the algorithm for face morphing attack detection.

A. Dataset Description

A publicly available biometric dataset is used in this paper, which combines synthetic facial images and a public morphing dataset. The synthetic morphing attack detection development (SMDD) dataset is utilized in this work [15]. It contains 500,000 synthetic facial images, which were generated by the special generative adversarial network (GAN) named StyleGAN2-ADA, trained on the FFHQ dataset and sampled from a Gaussian noise distribution. Image quality was assessed using the CR-FIQA technique, resulting in high-quality images. These were split into 25,000 bona fide samples and 25,000 probe images for morph generation. Morphing attacks were created using a landmark-based morphing approach with OpenCV and dlib, and a subsequent manual inspection removed samples with visible artifacts, yielding a final balanced subset of 15,000 bona fide and 15,000 morph images. An equivalent number of samples from the FRLL-Morphs dataset was incorporated following the same protocol to ensure dataset consistency and balanced class distribution.

B. Data Pre-processing

All facial images were first resized and normalized to ensure a consistent input resolution. From the normalized RGB images, multiple handcrafted texture representations were extracted, including local binary patterns (LBP) computed per color channel, histogram of oriented gradients (HOG) to emphasize edge and structural information, and binarized statistical image features (BSIF) to capture learned texture patterns. These feature maps were stacked to form a multi-channel input representation. During model inference,

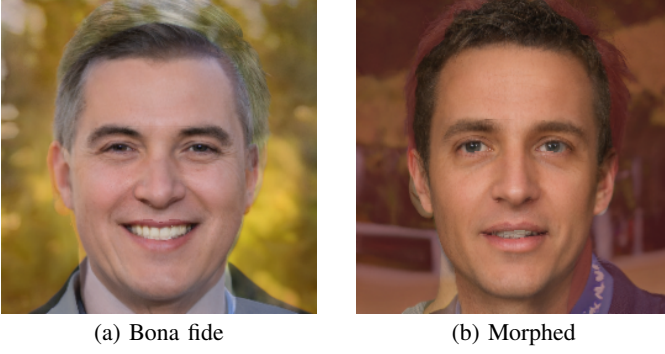


Fig. 1: Bona fide and morphed facial samples.

reconstruction-based residual maps were obtained from both Xception and ResNet50 decoder branches, highlighting subtle texture inconsistencies between bona fide and morph images. This pre-processing pipeline enhances morphing artifacts before classification.



Fig. 2: Data Pre-processing

C. Feature Extraction

To enhance sensitivity to morphing artifacts, multiple handcrafted texture descriptors are extracted, including RGB-based Local Binary Patterns (LBP), Histogram of Oriented Gradients (HOG), and Binarized Statistical Image Features (BSIF). These descriptors emphasize local texture inconsistencies commonly introduced during face morphing.

D. Dual-stream Residual Fusion Network (DRF-Net)

To implement the proposed framework, this research introduces a dual-stream residual fusion network (DRF-Net) designed to learn semantic facial features and morphing-induced residual artifacts jointly. DRF-Net consists of two parallel encoder-decoder streams as shown by the figure 3 that operate on the handcrafted feature-enhanced input representation. One stream employs an Xception-based encoder to capture fine-grained texture-related patterns, while the second stream utilizes a ResNet50-based encoder to extract robust

hierarchical and semantic features. In each stream, encoder outputs are passed through a U-Net-style decoder to reconstruct the input representation. Residual maps are obtained by subtracting the reconstructed outputs from the corresponding inputs, explicitly modeling inconsistencies introduced during the morphing process. Global Average Pooling is applied to both encoder bottleneck features and residual maps to generate compact semantic and residual feature vectors. These features are fused through concatenation and passed to a fully connected classifier for final decision making.

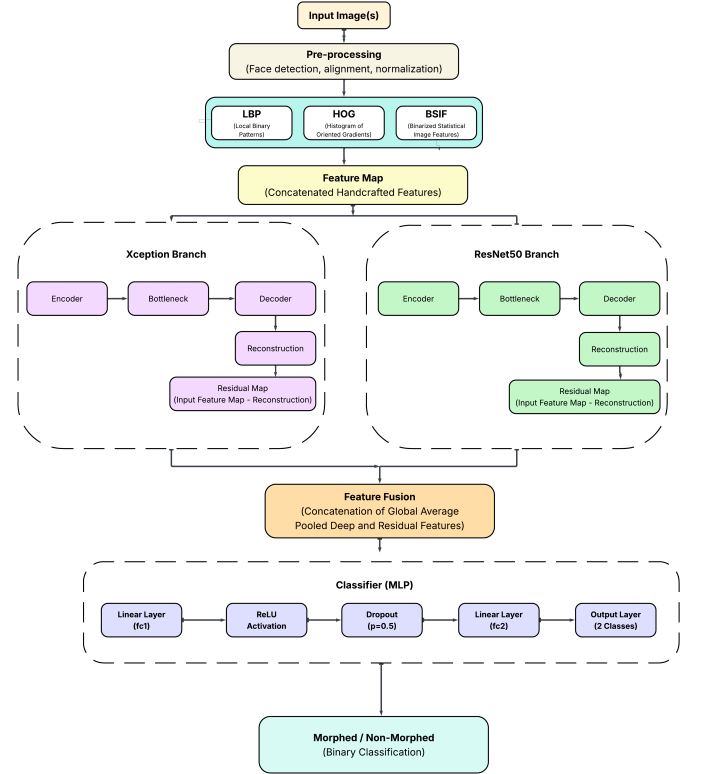


Fig. 3: The proposed DRF Net architecture

1) *UNet-Xception Reconstruction*: The UNet-Xception is designed as an encoder-decoder reconstruction module. It receives the concatenated feature map as input and processes it through an Xception-based encoder. The encoder uses depth-wise separable convolutional operations to extract hierarchical feature representations while maintaining computational efficiency. Through progressive downsampling, the encoder captures features at multiple abstraction levels and produces a compact bottleneck representation. The decoder follows a UNet like upsampling structure to reconstruct the input feature representation from the bottleneck features. Skip connections are used between the corresponding encoder and decoder stages to preserve fine-grained spatial and texture information. The final output of the decoder represents the reconstructed feature map of the UNet-Xception branch. This reconstructed output is later compared with the original input feature map to generate a residual map, which emphasizes reconstruction errors and possible morphing artifacts.

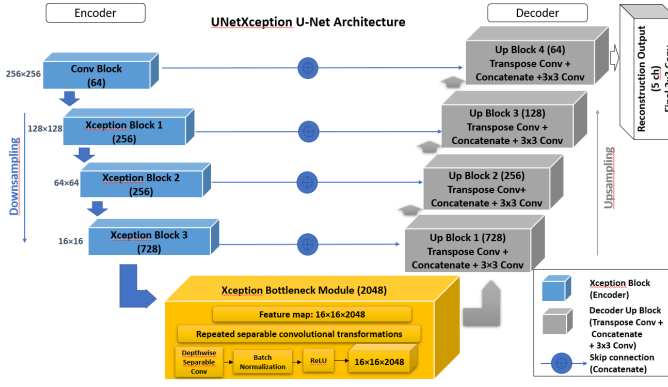


Fig. 4: The UnetXception architecture

2) *UNet-ResNet Reconstruction*: In parallel with the UNet-Xception, the UNet-ResNet is used to learn complementary residual representations. The ResNet-based encoder extracts deep residual features using residual learning blocks, which help preserve important discriminative information and reduce feature degradation during deep feature extraction. Similarly to the Xception, the encoder progressively reduces the spatial resolution while increasing the representational capacity of the feature maps. The decoder reconstructs the input feature representation using upsampling operations and skip connections from the corresponding encoder stages. These skip connections help restore spatial details that may be lost during downsampling. The reconstructed feature map generated by the UNet-ResNet branch is compared with the original input feature map to produce another residual map. This branch provides complementary reconstruction behavior, allowing the framework to capture morphing artifacts from a different deep feature perspective.

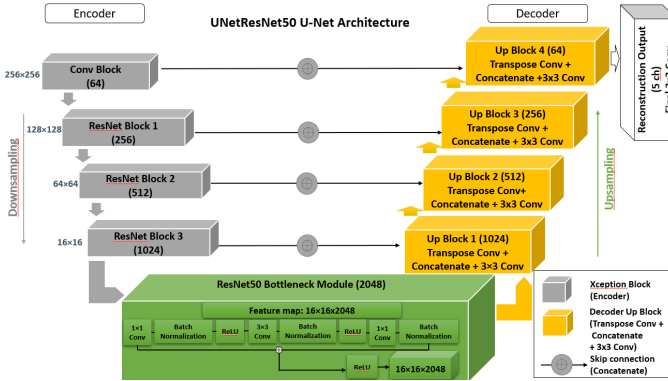


Fig. 5: The UnetResnet50 architecture

3) *Residual Map Generation and Feature Fusion*: Residual maps are generated by comparing the original handcrafted feature map with the reconstructed outputs of both reconstruction branches. The residual map highlights the difference between the input feature representation and its reconstructed version. Since morphed facial images may contain subtle blending artifacts and texture inconsistencies, reconstruction errors can

provide useful cues for detecting such attacks. For feature fusion, global average pooling is applied to bottleneck features and residual maps obtained from both UNet-Xception and UNet-ResNet branches. The pooled feature vectors are then concatenated to form a single fused feature representation. This fused vector combines deep bottleneck-level information with residual artifact-based features, enabling the classifier to make a more robust final decision.

4) *Classification Stage*: In the classification stage, the fused feature vector obtained from the feature fusion module is passed into a Multi-Layer Perceptron (MLP) classifier for final prediction. The MLP classifier consists of a linear layer, a ReLU activation function, a dropout layer, and a final linear output layer. The first linear layer learns discriminative patterns from the fused feature representation, while the ReLU activation introduces non-linearity to capture complex relationships among the extracted features. The dropout layer is applied to reduce overfitting and improve the generalization capability of the model. Finally, the last linear layer generates class scores for binary classification, where each input face image is classified as either morphed or non-morphed. During training, cross-entropy loss is used to optimize the classifier and enhance its ability to distinguish morphing artifacts from genuine facial patterns.

Algorithm 1 DRF-Net: Part-wise Dual-stream Residual Fusion Model

Input face image I Predicted class
 $\hat{y} \in \{\text{bona fide, morph}\}$

Part I: Preprocessing and Handcrafted Feature Extraction

$I \leftarrow \text{Preprocess}(I)$; $F_R \leftarrow \text{LBP}_R(I)$; $F_G \leftarrow \text{LBP}_G(I)$;
 $F_B \leftarrow \text{LBP}_B(I)$; $F_H \leftarrow \text{HOG}(I)$; $F_S \leftarrow \text{BSIF}(I)$;
 $X \leftarrow \text{Concat}(F_R, F_G, F_B, F_H, F_S)$

Part II: Dual-stream Deep Feature Encoding

$\{Z_X^l\}_{l=1}^L \leftarrow \mathcal{E}_{Xcep}(X)$
 $\{Z_R^l\}_{l=1}^L \leftarrow \mathcal{E}_{Res50}(X)$

Part III: Decoder-based Reconstruction

$\hat{X}_X \leftarrow \mathcal{D}_{Xcep}(\{Z_X^l\}_{l=1}^L)$
 $\hat{X}_R \leftarrow \mathcal{D}_{Res50}(\{Z_R^l\}_{l=1}^L)$

Part IV: Residual Noise Computation

$R_X \leftarrow X - \hat{X}_X$
 $R_R \leftarrow X - \hat{X}_R$

Part V: Semantic and Residual Feature Pooling

$s_X \leftarrow \text{GAP}(Z_X^L)$
 $s_R \leftarrow \text{GAP}(Z_R^L)$
 $r_X \leftarrow \text{GAP}(R_X)$
 $r_R \leftarrow \text{GAP}(R_R)$

Part VI: Feature Fusion and Classification

$f \leftarrow \text{Concat}(s_X, r_X, s_R, r_R)$
 $\hat{y} \leftarrow \text{Softmax}(\text{FC}(f))$

IV. RESULTS AND DISCUSSION

A. Performance Evaluation of the proposed Model

The performance of the proposed DRF-Net model was evaluated using standard metrics derived from the confusion matrix shown in Figure 6. Out of 3,200 test samples, the model correctly classified 1,575 bona fide images and 1,583 morphing attacks, with only 25 bona fide samples misclassified as morphs and 17 morph samples misclassified as bona fide.

This results in an overall classification accuracy of 98.69%. Considering morphing attacks as the positive class, the model achieved a precision of 98.45% and a recall (true positive rate) of 98.94%, indicating strong detection capability with minimal false alarms. The corresponding F1-score was 98.69%, reflecting a balanced trade-off between precision and recall. In terms of biometric security metrics, the Attack Presentation Classification Error Rate (APCER) was 1.06%, while the Bona Fide Presentation Classification Error Rate (BPCER) was 1.56%, resulting in an Average Classification Error Rate (ACER) of 1.31%. These results demonstrate the robustness and reliability of the proposed approach for face morphing attack detection.

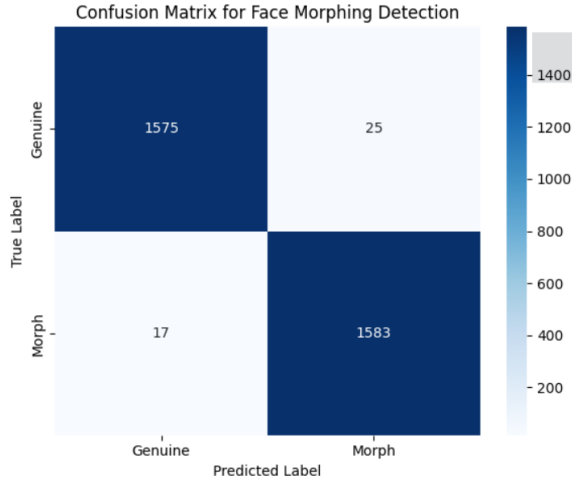


Fig. 6: Confusion Matrices

Figure 7 shows the training and validation loss and accuracy curves of the proposed DRF-Net model. The loss curves exhibit stable convergence, while the accuracy curves demonstrate consistent improvement and close alignment between training and validation phases. The absence of significant divergence indicates effective learning and good generalization. These trends highlight the robustness of the proposed dual-stream residual fusion architecture.

B. Comparison with SOTA Pre-trained Models

Table I presents a comparative evaluation of different deep learning architectures for face morphing attack detection. The results show that encoder-decoder-based models consistently achieve strong performance, with improvements observed when deeper backbones are employed. Lightweight architectures also demonstrate competitive results, highlighting

their suitability for efficient deployment. Overall, the comparison confirms the effectiveness of reconstruction-based and residual-aware designs for morphing attack detection.

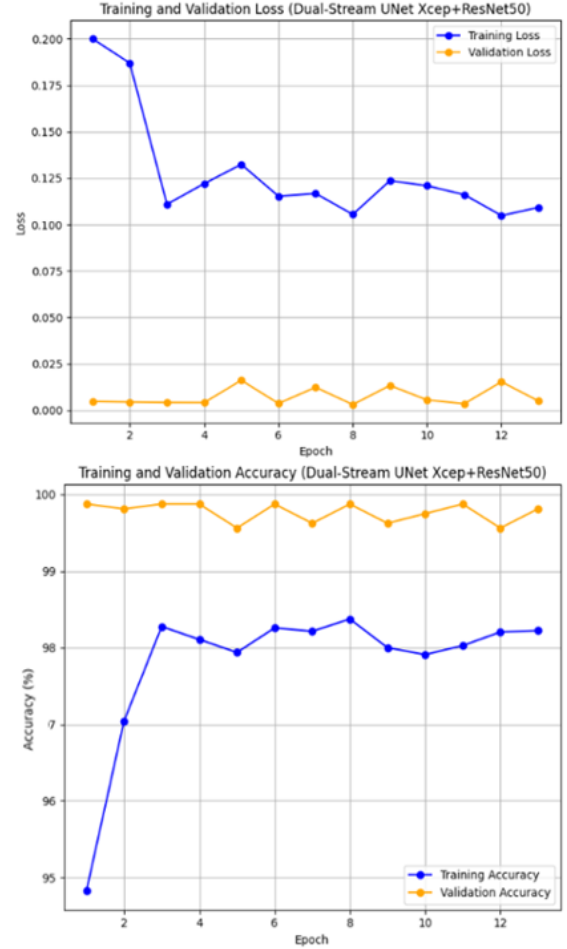


Fig. 7: Confusion Matrices

TABLE I: Performance Comparison of Different Models for Face Morphing Detection

Model	Acc.	Prec.	Rec.	F1	BPCER	EER
UNetResNet18	96.88	0.9706	0.9688	0.9687	0.0625	0.0625
UNetResNet50	96.81	0.9700	0.9681	0.9681	0.0638	0.0612
DenseNet121	96.09	0.9631	0.9609	0.9609	0.0731	0.0669
DenseNet121+Dec	96.09	0.9631	0.9609	0.9609	0.0731	0.0669
MobileNetV2	97.94	0.9802	0.9794	0.9794	0.0413	0.0088
DRF-Net	98.69	0.9869	0.9869	0.9869	0.0156	0.0119

C. Comparison with Previous Work

Table II presents a comparative analysis between the proposed DRF-Net and existing state-of-the-art face morphing detection methods using BPCER and EER as evaluation metrics. The UNetXception method [16] published in 2025 achieves a BPCER of 0.44 and EER of 0.31, while the D-FMAD [17] shows better performance of 0.11 BPCER and 0.025 EER. The MS-CAN [18] approach reports a BPCER of 0.0523 and an EER of 0.0418, indicating moderate performance in handling

morphing attacks. The P-LBP De-DCNN method [9] achieves improved results with a BPCER of 0.0485 and an EER of 0.0335, demonstrating the effectiveness of deep CNN-based texture representations. In contrast, the proposed DRF-Net significantly outperforms all competing methods, achieving the lowest BPCER of 0.0156 and EER of 0.0119. This substantial reduction in error rates highlights the robustness and reliability of the proposed approach in accurately detecting face morphing attacks while minimizing both bona fide and attack classification errors. The results clearly demonstrate the superiority of DRF-Net over existing state-of-the-art methods.

TABLE II: Comparison with State-of-the-Art Face Morphing Detection Methods

Work	Method	BPCER	EER
Dhore et al. (2025) [16]	UNetXception	0.44	0.31
Hoang et al. (2025) [17]	D-FMAD	0.11	0.025
Venkatesh et al. (2019) [18]	MS-CAN Residual Noise	0.0523	0.0418
Raja et al. (2022) [9]	P-LBP De-DCNN	0.0485	0.0335
Proposed Model	DRF-Net	0.0156	0.0119

V. CONCLUSION

This paper addressed the problem of face morphing attack detection in biometric face recognition systems by proposing DRF-Net, a dual-stream residual fusion framework that integrates deep learning with reconstruction-based residual analysis. By jointly learning semantic facial representations and residual noise artifacts through parallel encoder-decoder architectures, the proposed method effectively captures subtle morphing inconsistencies that are difficult to detect in the spatial domain. Experimental results demonstrate that DRF-Net achieves consistently low error rates, including a BPCER of 0.0156 and an EER of 0.0119, outperforming existing state-of-the-art methods under the same experimental setting. These results confirm the robustness and reliability of the proposed approach for high-security biometric applications. Future work will focus on cross-dataset generalization and extending the framework to video-based and real-time morphing attack detection scenarios.

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